

1.

```
section .data
    msg db '1','2','3','4','5','6','7','8','9', 10
    len equ $ - msg

section .text
    global _start

_start:
    mov eax, 4          ; sys_write
    mov ebx, 1          ; stdout
    mov ecx, msg
    mov edx, len
    int 0x80

    mov eax, 1          ; sys_exit
    int 0x80
```

STDIN

Input for the program ( Optional )

Output:

1 m

123456789

2.

```
section .bss
    result resb 4

section .text
    global _start

_start:
    mov ecx, 5
    mov edx, 10

    call sum          ; result in EAX

    ; convert number to ASCII (simple for 1-99)
    add eax, '0'      ; convert to character
    mov [result], al

    ; print result
    mov eax, 4
    mov ebx, 1
    mov ecx, result
    mov edx, 1
    int 0x80

    ; exit
    mov eax, 1
    int 0x80

sum:
    mov eax, ecx
    add eax, edx
    ret
```

STDIN

Input for the program ( Optional )

Output:

1 ms | 4.5 MB

?

3.

|                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <pre>section .bss     char resb 1  section .text     global _start  _start:     call display      mov eax, 1     int 0x80  display:     mov ecx, 0        ; counter  print_loop:     mov [char], cl      mov eax, 4     mov ebx, 1     mov ecx, char     mov edx, 1     int 0x80      inc cl     cmp cl, 255     jne print_loop  ret</pre> | <p>STDIN</p> <p>Input for the program ( Optional )</p> <p>Output: 64 ms   4.5 MB</p> <p>Output limit exceeded (max 65536 bytes)</p> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|

4.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>%macro print_msg 2     mov eax, 4     mov ebx, 1     mov ecx, %1     mov edx, %2     int 0x80 %endmacro  section .data     msg1 db "Hello Programmers!", 10     len1 equ \$ - msg1      msg2 db "Welcome to the world of,", 10     len2 equ \$ - msg2      msg3 db "Linux Assembly programming!", 10     len3 equ \$ - msg3  section .text     global _start  _start:     print_msg msg1, len1     print_msg msg2, len2     print_msg msg3, len3      mov eax, 1     int 0x80</pre> | <p>STDIN</p> <p>Input for the program ( Optional )</p> <p>Output:</p> <p>Hello Programmers!<br/>Welcome to the world of,<br/>Linux Assembly programming!</p> |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|

5.

```
section .data
    msg db 'Looping...', 10
    len equ $ - msg

section .text
    global _start

_start:
    mov ecx, 5            ; loop 5 times

loop_start:
    mov eax, 4
    mov ebx, 1
    mov ecx, msg
    mov edx, len
    int 0x80

    loop loop_start      ; automatically decrements ECX

    mov eax, 1
    int 0x80
```

STDIN

Input for the program ( Optional )

Output:

7 ms | 4.6 MB

[illegible]